GEOGRAPHY • RAS PRELIMS NOTES

Geography of India

Physiographic divisions, climate & monsoon, rivers & lakes, irrigation, agriculture, minerals and industries

Inside these notes

- Every syllabus topic, explained simply and connected
- River, crop, mineral and industry tables, the way RAS asks them
- Memory hooks and exam traps in every chapter

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Notes made for RAS aspirants - simple language, connected concepts, PYQ-based facts.

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How to use: read a chapter fully once, then revise only the tables, MEMORY HOOKS and EXAM TRAPS. Every fact here is written the way RAS papers ask it.

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India: Location, Extent and Frontiers

India is a vast country lying entirely in the **Northern Hemisphere**. The mainland extends between latitudes **8°4′N and 37°6′N** and longitudes **68°7′E and 97°25′E**. The Tropic of Cancer (23°30′N) divides the country into almost two equal parts — the tropical south and the sub-tropical north.

- **Total area:** 3.28 million sq km — India is the **7th largest** country in the world (after Russia, Canada, China, USA, Brazil, Australia).
- **Land frontier:** about 15,200 km. **Coastline:** 7,516.6 km including islands (mainland coastline is about 6,100 km).
- East-west extent is about 2,933 km; north-south extent is about 3,214 km — yet the local time difference between Gujarat and Arunachal Pradesh is nearly two hours.
- **Indian Standard Time** is taken from the **82°30′E** meridian, which passes through Mirzapur (Uttar Pradesh) and through five states: UP, Madhya Pradesh, Chhattisgarh, Odisha and Andhra Pradesh. India is 5 hours 30 minutes ahead of GMT.
- Largest state by area: **Rajasthan**. Smallest state: **Goa**. Largest Union Territory: Ladakh. Largest district: Kachchh (Gujarat).
- Longest coastline among states: **Gujarat** (longest mainland coastline). Among UTs, the Andaman and Nicobar Islands have the longest coastline.

MEMORY HOOK: The Tropic of Cancer passes through **8 states** — remember west to east: **Gu**jarat, **Ra**jasthan, **M**adhya Pradesh, **Chha**ttisgarh, **Jha**rkhand, **W**est Bengal, **Tri**pura, **Mi**zoram ("Gu-Ra-Ma-Chha-Jha-We-Tri-Mi").

Land borders

India shares its land border with **seven countries**. Bangladesh shares the **longest** border with India; Afghanistan the shortest (with the Wakhan corridor region of Ladakh).

Country	Border length (km)	Indian states/UTs touching it
Bangladesh	4,096	West Bengal, Assam, Meghalaya, Tripura, Mizoram
China	3,488	Ladakh, Himachal Pradesh, Uttarakhand, Sikkim, Arunachal Pradesh
Pakistan	3,323	Ladakh, Jammu & Kashmir, Punjab, Rajasthan, Gujarat
Nepal	1,751	Uttarakhand, UP, Bihar, West Bengal, Sikkim
Myanmar	1,643	Arunachal Pradesh, Nagaland, Manipur, Mizoram
Bhutan	699	Sikkim, West Bengal, Assam, Arunachal Pradesh
Afghanistan	106	Ladakh (PoK side)

EXAM TRAP: The **longest** land border is with Bangladesh (4,096 km), NOT China. The McMahon Line is associated with the India-China border in the east (Arunachal sector); the Radcliffe Line divided India and Pakistan; the Durand Line is between Pakistan and Afghanistan.

PYQ PATTERN: RAS repeatedly asks the descending order of border lengths and "which state touches the maximum number of neighbouring states" — Uttar Pradesh touches the maximum number of states/UTs (9 including Delhi) plus Nepal.

Physiographic Divisions of India

India's physical features are very diverse because the landmass was built in different geological eras: the ancient Peninsular block (part of Gondwanaland), the young fold mountains of the Himalayas (formed by the collision of the Indo-Australian plate with the Eurasian plate), and the recent alluvial plains deposited between them.

On the basis of relief, India is divided into **six physiographic divisions**: (1) The Northern and North-Eastern Mountains, (2) The Northern Plains, (3) The Peninsular Plateau, (4) The Indian Desert, (5) The Coastal Plains, (6) The Islands.

MEMORY HOOK: Remember the division logic — Old plateau in the south, Young mountains in the north, New plains in between, Desert in the west, Coasts on both sides, Islands in both seas.

The Himalayas

The Himalayas are young fold mountains forming an arc about **2,400 km** long, running from the Indus bend (Nanga Parbat) in the west to the Brahmaputra bend (Namcha Barwa) in the east. Their width varies from about **400 km in Kashmir to 150 km in Arunachal Pradesh**. They rise steeply like a wall and protect India from cold Central Asian winds while forcing the monsoon to shed rain.

The three parallel ranges (north to south)

Range	Other names	Average height	Key facts
Himadri	Greater Himalaya, Inner Himalaya	~6,000 m	Highest and most continuous range; core of granite; perennially snow-covered; source of great rivers
Himachal	Lesser Himalaya, Middle Himalaya	3,700–4,500 m	Rugged; Pir Panjal, Dhauladhar, Nag Tibba, Mahabharat ranges; famous hill stations
Shiwalik	Outer Himalaya	900–1,100 m	Outermost range, 10–50 km wide, made of unconsolidated river sediments; cut by wide valleys called **Duns**

- **Mount Everest** (8,848.86 m, Nepal) is the world's highest peak. **Kanchenjunga** (8,598 m, Sikkim) is the highest peak **within India**. **K2 / Godwin-Austen** (8,611 m) lies in the **Karakoram** range, in the Gilgit-Baltistan region under Pakistan's occupation but claimed by India — it is the world's second-highest peak.
- Other great Himadri peaks: Dhaulagiri and Annapurna and Makalu (Nepal), Nanda Devi (Uttarakhand, highest peak entirely within India at 7,816 m), Namcha Barwa (Arunachal side, east), Nanga Parbat (western anchor).
- The Himachal range carries the hill stations: Shimla, Mussoorie, Nainital, Darjeeling, Ranikhet, Kasauli.

- In the east, the Shiwalik valleys are called **Duars** (e.g., Bengal Duars — tea gardens); in the west they are called **Duns** (Dehradun is the largest dun).

EXAM TRAP: **Nanda Devi** (7,816 m, Uttarakhand) is the highest peak *entirely* inside India. Kanchenjunga is higher but sits on the India–Nepal border. K2 is NOT in the Himalaya proper — it is in the **Karakoram**.

Trans-Himalaya (north of the Himadri)

North of the Greater Himalaya lie the Trans-Himalayan ranges in Ladakh: the **Karakoram** (with K2), the **Ladakh range**, and the **Zaskar range** (the Indus flows between Ladakh and Zaskar ranges). The **Siachen glacier** in the Karakoram is one of the largest glaciers outside the polar regions. This is a cold desert region lying in the rain shadow.

Regional divisions of the Himalaya (west to east)

The Himalaya is divided regionally by the river valleys that cut across it:

Division	Between the rivers	Main areas
Punjab/Kashmir-Himachal Himalaya	Indus and Satluj	Kashmir, Himachal; Karewa formations famous for saffron (Pampore)
Kumaun Himalaya	Satluj and Kali	Uttarakhand; Nanda Devi, Kamet, Badrinath, Kedarnath
Nepal Himalaya	Kali and Teesta	Longest stretch; Everest, Kanchenjunga, Dhaulagiri, Annapurna
Assam Himalaya	Teesta and Dihang (Brahmaputra)	Sikkim, Bhutan, Arunachal; Namcha Barwa

MEMORY HOOK: River boundaries west to east — **I**ndus, **S**atluj, **K**ali, **T**eesta, **D**ihang: "I Sat Kali Teesta Dihang".

Important Himalayan passes

Pass	Connects / located in	Note
Banihal	Jammu – Srinagar (J&K)	Jawahar Tunnel passes under it; NH-44
Pir Panjal	Jammu – Kashmir valley	Historic pass of the Pir Panjal range
Zoji La	Srinagar – Leh	On the Srinagar–Leh highway
Khardung La	Leh – Nubra valley (Ladakh)	Among the highest motorable passes in the world
Rohtang	Manali – Lahaul (Himachal)	Atal Tunnel (9.02 km) runs below it
Bara Lacha La	Himachal – Ladakh	Source region of Chenab's Chandra and Bhaga streams
Shipki La	Himachal – Tibet	The Satluj enters India through this pass
Mana and Niti	Uttarakhand – Tibet	Near Badrinath region
Lipulekh	Uttarakhand – Tibet	Route of the Kailash Mansarovar Yatra
Nathu La	Sikkim – Tibet (China)	India–China border trade route; reopened in 2006

Pass	Connects / located in	Note
Jelep La	Sikkim – Tibet	Lies east of Nathu La
Bomdi La	Arunachal Pradesh	On the way to Tawang
Bum La	Arunachal – Tibet	Near Tawang

EXAM TRAP: Shipki La is where the **Satluj** enters India, Nathu La is in **Sikkim** (not Arunachal), and Lipulekh is the Kailash Yatra pass in **Uttarakhand**.

Purvanchal — the north-eastern hills

South of the Brahmaputra valley, the Himalaya bends sharply at the Dihang gorge and runs south as a series of low hills called the **Purvanchal**: **Patkai Bum** (Arunachal–Myanmar border), **Naga Hills** (highest point Mt. Saramati), **Manipur hills**, and **Mizo (Lushai) hills**. West of these lie the **Garo, Khasi and Jaintia hills** of Meghalaya — actually an extension of the Peninsular plateau separated from it by the Malda gap. Shillong Peak is the highest point of the Meghalaya plateau.

MEMORY HOOK: Purvanchal west-to-east and north-to-south order asked in PYQs — Patkai, Naga, Manipur, Mizo. Meghalaya hills run **Garo → Khasi → Jaintia** from west to east.

The Northern Plains

The Northern Plains were formed by the alluvial deposits of three great river systems — the **Indus, the Ganga and the Brahmaputra** — filling a deep depression between the Himalayas and the Peninsula over millions of years. They stretch about 2,400 km from Punjab to Assam, are 240–320 km wide, and cover roughly 7 lakh sq km. Being flat, fertile and well-watered, they are among the most densely populated and intensively farmed regions of the world.

The three sections of the plain

The **Punjab-Haryana plain** (Indus system, doabs between rivers), the **Ganga plain** (largest part; upper, middle and lower sections from UP to West Bengal), and the **Brahmaputra plain** (Assam valley, narrow and flood-prone). A **doab** is the fertile land between two rivers — the Bist doab (Beas–Satluj) and Bari doab (Beas–Ravi) are classic examples.

Relief divisions (north to south, across the plain)

Division	Meaning	Key facts
Bhabar	Pebble zone at Shiwalik foot	8–16 km wide belt of porous gravel; small streams **disappear** underground here
Terai	Marshy zone south of Bhabar	Streams **re-emerge**; wet, forested, rice-growing belt
Bangar	Old alluvium upland	Terraces above flood level; contains calcium nodules called **kankar**
Khadar	New alluvium lowland	Flood plains renewed by floods almost every year; most fertile

EXAM TRAP: Rivers disappear in the **Bhabar** and reappear in the **Terai** — never the reverse. Khadar = NEW alluvium (flooded yearly), Bangar = OLD alluvium (safe from floods).

PYQ PATTERN: The **Ganga–Brahmaputra delta** (Sundarbans, named after the Sundari tree) is the **largest delta in the world**. It lies mostly in Bangladesh with its western part (moribund delta) in West Bengal.

The Peninsular Plateau

The Peninsula is India's oldest landmass — a part of ancient **Gondwanaland**, made of hard igneous and metamorphic rocks, and stable (earthquake-resistant by comparison). Rivers here are old, graded, and flow in wide shallow valleys. The plateau is roughly triangular, sloping generally from west to east, which is why most peninsular rivers flow eastwards into the Bay of Bengal.

Its average elevation is 600–900 m. The plateau is divided by the Narmada rift valley into the **Central Highlands** (north) and the **Deccan Plateau** (south).

Central Highlands and its parts

- **Malwa Plateau** (MP) — lava-covered (black cotton soil), drained by Chambal, Betwa, Sind and Ken.
- **Bundelkhand** (UP–MP) — old, eroded granite-gneiss upland.
- **Baghelkhand** (MP–Chhattisgarh).
- **Chotanagpur Plateau** (Jharkhand) — India's richest mineral belt; the "storehouse of minerals"; drained and dissected by the Damodar river (Damodar valley is called the **Ruhr of India** for its coal and industry).
- **Aravalli Range** — one of the **oldest fold mountains in the world**, running from Gujarat to Delhi (through Rajasthan); heavily eroded relic range. Highest peak: **Guru Shikhar** (1,722 m, Mt. Abu).
- **Vindhya Range** — runs along the north bank of the Narmada; traditional boundary between North and South India; source of Chambal, Betwa, Ken, Son headstreams.
- **Satpura Range** — block mountains south of the Narmada, between Narmada and Tapi rift valleys; highest peak **Dhupgarh** (1,350 m, near Pachmarhi, MP); Mahadeo and Maikal hills; **Amarkantak** (Maikal) is the source of the Narmada and the Son — rivers flowing in opposite directions from one hill.

MEMORY HOOK: Between two rift valleys — **Narmada** in the north, **Tapi** in the south — sits the **Satpura**. Vindhya guards Narmada's north bank; Satpura its south.

Deccan Plateau and the Ghats

The **Deccan Plateau** covers Maharashtra, Karnataka, Telangana, Andhra Pradesh and Tamil Nadu uplands. Its western edge is the **Western Ghats** and its eastern edge the broken **Eastern Ghats**.

Feature	Western Ghats (Sahyadri)	Eastern Ghats
Continuity	Continuous wall, 1,600 km long	Discontinuous, broken by rivers
Average height	900–1,600 m, rises southward	About 600 m
Highest peak	**Anamudi** (2,695 m, Kerala) — highest in South India	**Mahendragiri** (1,501 m, Odisha) as per NCERT
Rivers	Short, swift streams; heavy west-side rainfall	Mahanadi, Godavari, Krishna, Kaveri cut across to the sea
Passes/gaps	Thal Ghat, Bhor Ghat, Pal Ghat	Hills: Nallamala, Palkonda, Javadi, Shevaroy

- **Nilgiri Hills** (Tamil Nadu) are where the Western and Eastern Ghats meet; **Doddabetta** (2,637 m) is the highest peak of the Nilgiris. Ooty (Udhagamandalam) sits in the Nilgiris.
- South of the Nilgiris lie the **Anamalai** (with Anamudi) and **Cardamom hills**. Between the Nilgiri and Anamalai ranges lies the wide **Pal Ghat (Palakkad) gap**, the main gateway between Kerala and Tamil Nadu.
- **Thal Ghat** connects Mumbai–Nashik; **Bhor Ghat** connects Mumbai–Pune.

EXAM TRAP: **Anamudi** (2,695 m) is the highest peak of Peninsular India AND the Western Ghats. Doddabetta is second. **Pal Ghat** is a gap/pass, not a peak. The Western Ghats are higher and more continuous than the Eastern Ghats — the reverse statement is a common wrong option.

PYQ PATTERN: Statement-style RAS questions say "the Eastern Ghats are higher than the Western Ghats" or "the Narmada flows between the Satpura and the Aravalli" — both are traps; Narmada flows between the **Vindhya** (north) and **Satpura** (south).

The Indian Desert

West of the Aravallis lies the **Thar Desert** — about 2 lakh sq km covering western Rajasthan and extending into Gujarat, Haryana, Punjab and across the border into Pakistan. It receives less than 150 mm of annual rainfall, has an arid climate and scanty vegetation of thorny bushes.

- Crescent-shaped sand dunes are called **barchans**; linear dunes run parallel to the wind. Shifting dunes near Jaisalmer are locally called **dhrian**.
- The **Luni** is the only significant river of the desert; it rises in the Aravallis near Ajmer and flows to the Rann of Kutch, turning **saline** after Balotra — it is an inland (ephemeral/endorheic) drainage.
- East of the desert, the semi-arid **Bagar** tract and the Shekhawati region form the transition to the plains.

The Coastal Plains

India's mainland coast is flanked by narrow plains that differ sharply between west and east because the west coast is a **submergent, high-energy coast** while the east coast is **emergent with big deltas**.

Feature	Western Coastal Plain	Eastern Coastal Plain
Width	Narrow (10–25 km in stretches)	Wider (100–120 km)
Subdivisions	Konkan (Maharashtra–Goa), Kannad (Karnataka), Malabar (Kerala)	Utkal (Odisha), Andhra coast, Coromandel (Tamil Nadu)
River mouths	Estuaries, no deltas	Large deltas of Mahanadi, Godavari, Krishna, Kaveri
Special features	Lagoons and backwaters (**kayals**) in Kerala	Natural harbours fewer; ports: Chennai, Paradip, Visakhapatnam, Tuticorin
Ports	Mumbai, Mormugao, Mangaluru, Kochi	Chennai, Paradip, Visakhapatnam, Ennore, Tuticorin

- **Vembanad** (Kerala) is the **longest lake** of India and the centre of the Kerala backwaters; the famous Nehru Trophy boat race is held in the Punnamada stretch of these backwaters in Alappuzha.
- **Lake Chilika** (Odisha) sits at the northern end of the east coast; **Lake Pulicat** lies on the Andhra–Tamil Nadu border; **Kolleru** lies between the Godavari and Krishna deltas.

MEMORY HOOK: West coast divisions north to south — **Konkan, Kannad, Malabar** ("Ko-Ka-Ma"). East coast north to south — **Utkal, Andhra, Coromandel**.

The Islands

Andaman and Nicobar Islands (Bay of Bengal)

A chain of about 572 islands (including islets), these are submerged extensions of submarine mountains (the Arakan Yoma continuation). Only about three dozen are inhabited. Port Blair is the capital.

- The **Ten Degree Channel** separates the Andaman group from the Nicobar group; the **Duncan Passage** separates South Andaman from Little Andaman; the **Great Channel** separates Great Nicobar from Sumatra (Indonesia).
- **Barren Island** is India's **only active volcano**; **Narcondam** is a dormant volcanic island — both are in the Andaman Sea.
- **Saddle Peak** (732 m) is the highest point of the islands (North Andaman). Indira Point on Great Nicobar is India's southernmost point.

Lakshadweep (Arabian Sea)

A group of **36 small coral islands/atolls** (about 32 sq km total — the **smallest UT** by area), 220–440 km off the Kerala coast; capital **Kavaratti**. They are built of coral deposits. **Minicoy** is the southernmost island; **Androtт** is the largest island.

- The **Nine Degree Channel** separates the main Lakshadweep group from Minicoy; the **Eight Degree Channel** separates Minicoy (India) from the Maldives.

EXAM TRAP: Channel ladder from south — **8° Channel** (India–Maldives), **9° Channel** (Lakshadweep–Minicoy), **10° Channel** (Andaman–Nicobar). Most students wrongly put the 10 Degree Channel between India and Sri Lanka — India and Sri Lanka are separated by the **Palk Strait** and the Gulf of Mannar.

Climate of India

India's climate is described as the **monsoon type**. The word monsoon comes from the Arabic word **"mausim"**, meaning season — it refers to the **seasonal reversal of winds**. The Arab scholar **Al-Masudi** gave an early description of the monsoon. In summer, winds blow from the sea to the land (wet); in winter, from the land to the sea (dry). India also lies in the tropical belt, moderated by a long coastline, so the climate is hot with relatively small annual temperature ranges in the south and large ranges in the interior north.

Factors controlling India's climate

- **Latitude** — the Tropic of Cancer cuts India in half; the south is truly tropical, the north sub-tropical.
- **The Himalayan wall** — blocks cold Central Asian winds in winter (keeping India warmer) and traps the monsoon winds in summer, forcing them to rain over India.
- **Altitude** — temperature falls about 1°C for every 166 m of height (why hill stations are cool).
- **Distance from the sea** — the coast has an equable (moderate) climate; the deep interior (Delhi, Rajasthan) has an extreme, continental climate.

- **Pressure and winds** — seasonal shifting of pressure belts and the Inter-Tropical Convergence Zone (ITCZ).
- **Jet streams** — fast upper-air winds that steer surface weather.
- **El Nino and the Indian Ocean Dipole** — Pacific and Indian Ocean sea-surface conditions that strengthen or weaken the monsoon.

How the monsoon works — step by step

Understand this mechanism and every monsoon question becomes easy:

1. In summer (May–June) the sun shifts north; the land of north-west India and the Tibetan Plateau heat up intensely, creating a deep **low pressure**. 2. The **ITCZ** (the low-pressure trough near the equator where trade winds meet) shifts north to about 20°–25°N over the Ganga plain — this position is called the **monsoon trough**. 3. Over the southern Indian Ocean (near Madagascar — the Mascarene High) pressure stays **high**. Air flows from high to low pressure: the **south-east trade winds** start crossing the Indian Ocean towards India. 4. After crossing the equator, these winds are deflected to their **right** by the Coriolis force (Ferrel's law) and become the **south-west monsoon** winds, full of moisture from the warm ocean. 5. The intense heating of the huge **Tibetan Plateau** in summer creates an upper-air outflow — the **tropical easterly jet stream** blows over the Indian peninsula, and the **subtropical westerly jet stream** (which blew south of the Himalayas all winter) shifts north of the Himalayas. This shift is a key trigger for the **onset** of the monsoon. 6. Meeting the triangular shape of India, the SW monsoon splits into two branches — the **Arabian Sea branch** and the **Bay of Bengal branch**.

PYQ PATTERN: Statement questions love this chain: "ITCZ shifts north in summer", "winds deflect right after the equator", "Tibetan Plateau heating helps onset". All three are correct together.

The two branches of the south-west monsoon

Feature	Arabian Sea branch	Bay of Bengal branch
First strike	West coast (Kerala, Konkan, Karnataka)	North-east India (deflected by Arakan hills towards the Ganga plain)
Rainfall	Very heavy on the windward (west) side of the Western Ghats; rain shadow in the Deccan behind the Ghats	Heavy in Meghalaya/Assam; decreases westwards up the Ganga plain
Onset date	Kerala around **1 June**; Mumbai about 10 June	Reaches Bengal/Assam in early June
Later merge	One stream crosses to Saurashtra/Kachchh; another passes up the Narmada–Tapi valleys to central India	Travels west along the Himalayan foothills, finally merging with the Arabian Sea stream over Punjab–Haryana
Special point	A third stream strikes Saurashtra and passes along the Aravalli — gives scanty rain to Rajasthan	Responsible for the wettest places on earth in Meghalaya

- The **onset (arrival)** of the monsoon in Kerala normally occurs on **1 June**; it covers the whole country by about **early to mid-July (about 8 July)**.
- The monsoon arrives with a sudden burst of heavy rain called the **"burst of the monsoon"**.
- **Withdrawal (retreat)** begins from western Rajasthan around **mid-September** and clears the peninsula by mid-October — withdrawal is much more gradual than onset.

- **Break in the monsoon** — rainless intervals of one or more weeks during the season, caused by the shifting of the monsoon trough to the Himalayan foothills (then the hills flood while the plains stay dry).

EXAM TRAP: Onset is sudden ("burst") but withdrawal is gradual. Also, the **Arabian Sea branch gives rain first** (Kerala coast), and Tamil Nadu remains **dry during the SW monsoon** — it lies parallel to the Bay branch's path and in the rain shadow for the Arabian branch.

The four seasons (IMD pattern)

Season	Months	Hallmarks
Cold weather (winter)	Mid-November to February	Clear skies, low humidity, NE trade winds blow land-to-sea; frost in the north
Hot weather (summer)	March to May	Rising heat, low pressure over the NW; hot dry **loo** winds
South-west monsoon (rainy)	June to September	About **75% of India's annual rainfall**
Retreating monsoon (post-monsoon)	October to November	Skies clear, **October heat** (high humidity + heat), cyclones in the Bay; NE monsoon rains begin on the Tamil Nadu coast

Winter details

In winter the **subtropical westerly jet stream** blows **south of the Himalayas** and steers in cyclonic disturbances from the Mediterranean region called **western disturbances**. They bring light winter rain (**mahawat**) to Punjab, Haryana, Delhi and western UP — small in amount but **very useful for the rabi wheat crop** — and snowfall in the Himalayas. Peninsular India has no well-marked winter; temperature stays mild.

Summer details — local storms

- **Loo** — hot, dry, dusty winds of the northern plains in May–June afternoons.
- **Mango showers** — pre-monsoon showers on the Kerala–Karnataka coast that help early ripening of mangoes.
- **Blossom showers** — Kerala showers that help coffee blossoms.
- **Nor'westers / Kalbaisakhi** — violent evening thunderstorms in West Bengal and Assam in April–May; in Assam they are called **Bardoisila**. They help tea, jute and rice.

MEMORY HOOK: Summer showers by use — **M**ango showers ripen **M**angoes, **B**lossom showers bloom coffee, **K**albaisakhi (Bengal) and **B**ardoisila (Assam) brew tea.

Rainfall distribution

India's average annual rainfall is about **118 cm**, but its distribution is extremely uneven — this unevenness itself is a favourite exam theme.

- **Heavy rainfall (over 200 cm):** the western side of the Western Ghats, the north-east (Meghalaya, Assam, Arunachal), the sub-Himalayan West Bengal and parts of the Andaman–Nicobar.
- **Medium (100–200 cm):** eastern Madhya Pradesh, Odisha, the middle Ganga valley, parts of Gujarat.

- **Low (50–100 cm):** western UP, Haryana, Punjab, eastern Rajasthan, interior Deccan (rain shadow).
- **Scanty (below 50 cm):** western Rajasthan (Thar — Jaisalmer region gets under 15 cm), the Ladakh cold desert, and parts of the interior Deccan.
- Rainfall **decreases from east to west** across the northern plains, and **from the coasts towards the interior** of the peninsula.
- **Mawsynram** (Meghalaya, Khasi hills) records the **highest average annual rainfall in the world** (about 1,141 cm); neighbouring **Cherrapunji (Sohra)** is close behind — both lie on the southern funnel-shaped slopes of the Khasi hills facing the Bay branch.
- The **Tamil Nadu (Coromandel) coast** gets most of its rain in **winter (October–December)** from the retreating/NE monsoon and cyclones — about half or more of Tamil Nadu's annual rain comes in this season.

EXAM TRAP: Highest rainfall place = **Mawsynram**, not Cherrapunji. Lowest rainfall region = **western Rajasthan** (among the plains) — the Ladakh cold desert is equally dry, so read the options carefully.

PYQ PATTERN: Ordering questions ask districts/regions by **descending rainfall**: Western Ghats/NE at the top, then the east coast and middle Ganga valley, then western UP–Haryana, and the Thar at the bottom.

Retreating / north-east monsoon and cyclones

During October–November the low pressure over north India weakens and is replaced by high pressure; winds begin blowing from the land towards the Bay of Bengal (north-east direction). Crossing the Bay, they pick up moisture and dump heavy rain on the **Coromandel coast (Tamil Nadu, coastal Andhra)**. The retreating season is also the main **cyclone season** of the Bay of Bengal: the deltas of the Godavari, Krishna and Kaveri and the Odisha–West Bengal coast are the most cyclone-hit areas. Historically the Bay of Bengal generates roughly **four times** as many cyclones as the Arabian Sea, though Arabian Sea cyclones have become stronger in recent years.

El Nino, La Nina and the Indian Ocean Dipole

- **El Nino** — a warm ocean current appearing off the **Peru coast** (eastern Pacific) every 2–5 years, replacing the cold Peruvian current. It weakens the Indian monsoon and is associated with **drought** in India (e.g., the weak monsoons of strong El Nino years).
- **La Nina** — unusually cold eastern Pacific waters; generally associated with a **good (above-normal) Indian monsoon**.
- **Indian Ocean Dipole (IOD)** — a difference in sea-surface temperature between the western and eastern Indian Ocean. A **positive IOD** (warmer west) generally helps the Indian monsoon and can partly offset an El Nino; a negative IOD hurts it.

MEMORY HOOK: For the monsoon's Pacific link — "El Nino = drought fear, La Nina = rain cheer; positive IOD lends a hand."

Koeppen's climate types of India (frequently matched)

Code	Climate type	Region
Amw	Monsoon with short dry winter	West coast south of Goa (Malabar)
Aw	Tropical savanna	Most of the Peninsular plateau south of the Tropic of Cancer

Code	Climate type	Region
As	Monsoon with dry summer	**Coromandel coast** (rain in winter)
BShw	Semi-arid steppe	Rain-shadow Deccan, parts of Haryana–Punjab, eastern Rajasthan margins
BWhw	Hot desert	Extreme western Rajasthan (Thar)
Cwg	Monsoon with dry winter (Gangetic type)	The Ganga plain, eastern Rajasthan, northern MP, north-east
Dfc	Cold humid winter with short summer	Arunachal Pradesh (eastern Himalaya)
E	Polar type	High Himalaya — J&K, Ladakh, Himachal, Uttarakhand above snowline

EXAM TRAP: In Koeppen's scheme for India, **As (Coromandel) is the odd one — dry in summer, wet in winter**; almost everywhere else 'w' means dry winter.

Major Rivers of India

Think of India's rivers as two families. The **Himalayan rivers** are fed by glaciers plus rain, so they are **perennial**, young, fast, and carry huge loads of silt through gorges and meanders. The **Peninsular rivers** are fed only by rain, so they are **seasonal**, old, graded, and flow in broad shallow valleys at a gentle speed. This single contrast explains most river questions.

Feature	Himalayan rivers	Peninsular rivers
Water flow	Perennial (snow + rain)	Seasonal (rain only)
Stage	Youthful, active erosion, gorges, V-valleys	Mature/old, graded profile
Course	Long, change course, big meanders and oxbow lakes	Shorter/fixed courses, shallow valleys
Mouth	Huge deltas (Ganga–Brahmaputra)	East-flowing: deltas; west-flowing: estuaries
Navigation/irrigation	Useful for irrigation and hydropower	Mostly irrigation and hydropower

The Indus system

The **Indus** rises from the Bokhar Chu glacier near **Lake Mansarovar (Kailash region, Tibet)** — its source is popularly called **Sengge Khabab**, "the lion's mouth". It flows north-west between the Ladakh and Zaskar ranges, cuts a spectacular gorge near Nanga Parbat, and enters Pakistan near Chilas. Its total length is about **2,880 km**, of which about 1,114 km flows in India. It finally falls into the Arabian Sea near Karachi.

Under the **Indus Waters Treaty (1960)**, the three **eastern rivers — Ravi, Beas and Satluj — were allocated to India**, and the three **western rivers — Indus, Jhelum and Chenab — mainly to Pakistan** (India may use them in limited ways).

Tributary	Source / course	Exam fact
Jhelum	Verinag spring (Kashmir); flows through Srinagar and **Wular Lake**	Ancient name **Vitasta**; Mangla dam in Pakistan

Tributary	Source / course	Exam fact
Chenab	Formed by **Chandra + Bhaga** streams meeting at Tandi, near Bara Lacha La	Ancient name **Chandrabhaga / Asikni**; largest tributary of the Indus by volume among Indian tributaries
Ravi	Bara Bangahal near Rohtang (Himachal)	Ancient name **Iravati / Parushni**; Thein (Ranjit Sagar) dam
Beas	**Beas Kund** near Rohtang Pass	Ancient name **Vipasa**; Pong dam; joins Satluj at Harike
Satluj	**Rakshastal** near Mansarovar (Tibet); enters India at **Shipki La**	Ancient name **Satudri**; **Bhakra dam** (Gobind Sagar lake) and Nathpa Jhakri project

MEMORY HOOK: East–west order of Punjab's rivers and the doabs between them — Jhelum, Chenab, Ravi, Beas, Satluj. The Indus tributaries' ancient names form the Sapt-Sindhu of the Rigveda.

EXAM TRAP: The **Chenab** is formed by Chandra and Bhaga — students often swap it with the Alaknanda–Bhagirathi confluence. The Satluj enters India through **Shipki La**, and Bhakra dam is on the **Satluj**, not the Beas (Pong dam is on the Beas).

The Ganga system

The Ganga is India's **longest river (2,525 km)**. It begins in Uttarakhand as two headstreams: the **Bhagirathi** (from the Gangotri glacier at Gaumukh) and the **Alaknanda** (from the Satopanth glacier near Badrinath). They meet at **Devprayag**, and the combined stream is called the Ganga from there. It leaves the hills at Haridwar and flows through UP, Bihar, Jharkhand (briefly) and West Bengal, entering Bangladesh as the **Padma**.

The Panch Prayag (Alaknanda's five confluences, downstream order)

Prayag	Confluence
Vishnuprayag	Dhauliganga + Alaknanda
Nandprayag	Nandakini + Alaknanda
Karnaprayag	Pindar + Alaknanda
Rudraprayag	Mandakini + Alaknanda
Devprayag	Bhagirathi + Alaknanda = **Ganga**

MEMORY HOOK: Prayags in flow order spell "**Vi-Na-Ka-Ru-De**" — Vishnu, Nanda, Karna, Rudra, Dev.

Ganga's tributaries

Bank	Tributaries	Key facts
Left (from the Himalayas/north)	Ramganga, Gomti, Ghaghara, Gandak, Kosi, Mahananda	Rain-and-snow fed, bring floods and huge silt
Right (mostly from the Peninsula/south)	Yamuna, Son, Punpun, Damodar	Yamuna is the **longest tributary** of the Ganga

- **Yamuna** — rises from the Yamunotri glacier; its important tributaries are the **Chambal** (longest of the Yamuna tributaries by length in common texts), **Sind, Betwa and Ken** from the Peninsula, and the **Tons** from the Himalaya (the Tons actually carries more water than the Yamuna at their meeting near Kalsi). Cities on the Yamuna: Delhi, Agra, Mathura.

- **Ghaghara** (called Karnali in Nepal) is the Ganga's **largest tributary by water discharge** — a classic trap against the Yamuna, which is longest.
- **Gandak** (Narayani in Nepal) — forms the Triveni; the Gandak barrage at Valmikinagar/Bhaisalotan serves Bihar and Nepal.
- **Kosi** — joins from Nepal with seven streams (Sapta Kosi); notorious for shifting its course westwards and devastating floods — the **"Sorrow of Bihar"**.
- **Son** — rises near Amarkantak, flows north through a rift-like valley, joins the Ganga near Patna; tributaries include the Rihand (Govind Ballabh Pant Sagar reservoir in UP is among India's largest artificial lakes).
- **Damodar** — rises in Chotanagpur, flows through a rift valley rich in coal, joins the Hooghly; once called the **"Sorrow of Bengal"** for its floods, it was tamed by the **Damodar Valley Corporation (DVC, 1948)** — India's first multipurpose river valley project, modelled on the Tennessee Valley Authority (USA).
- **Farakka Barrage** (West Bengal) diverts Ganga water into the Bhagirathi–Hooghly channel to flush silt from the Kolkata port.

PYQ PATTERN: Match-the-following sets love these pairs — Kosi: Sorrow of Bihar; Damodar: Sorrow of Bengal; Brahmaputra: Sorrow of Assam; Mahanadi floods: Odisha. And "largest tributary by volume = Ghaghara" versus "longest tributary = Yamuna".

The Brahmaputra system

The **Brahmaputra** rises from the **Chemayungdung glacier** in the Kailash range near Mansarovar — very close to the sources of the Indus and the Satluj. In Tibet it is called the **Yarlung Tsangpo/Zangbo** and flows east for about 1,200 km. It takes a hairpin turn around **Namcha Barwa**, cuts a deep gorge, and enters Arunachal Pradesh as the **Siang/Dihang**. After joining the Dibang and the Lohit near Sadiya, it takes the name Brahmaputra in Assam. Its total length is about **2,900 km** (about 916 km in India).

- Major tributaries: **Subansiri, Kameng, Manas, Sankosh** (right bank, from Bhutan/Himalaya) and **Teesta** (from Sikkim, joins in Bangladesh); **Lohit, Dibang, Dhansiri, Kopili** (left bank).
- In Bangladesh it is called the **Jamuna**; after meeting the Ganga (Padma) their combined flow is the Padma, which then meets the Meghna and drains through the world's largest delta.
- **Majuli** (Assam) in the Brahmaputra is the **world's largest river island** and India's first island district. **Umananda** in Guwahati is one of the smallest river islands.
- The river is notorious for devastating annual floods and bank erosion in Assam — the **"Sorrow of Assam"**.

EXAM TRAP: The Brahmaputra is **longer than the Ganga in total length (2,900 vs 2,525 km)** and carries more water, but the **Ganga is called the longest river of India** because the greater part of the Ganga flows within India. The Brahmaputra's Tibet name is Tsangpo, Assam name is Brahmaputra, Bangladesh name is Jamuna.

Peninsular rivers — east-flowing (Bay of Bengal, delta-forming)

River	Length	Source	Key facts / projects
Mahanadi	851 km	Sihawa (Chhattisgarh)	**Hirakud dam** (Odisha) — one of the longest dams in the world; large delta; tributaries: Seonath, Hasdeo, Ib, Tel, Ong

River	Length	Source	Key facts / projects
Godavari	1,465 km	Trimbakeshwar (Nashik, Maharashtra)	**Longest peninsular river — "Dakshin Ganga"**; largest peninsular basin; tributaries: Pranhita (largest), Penganga, Wainganga, Wardha, Indravati, Sabari, Manjira; Polavaram project; Jayakwadi dam
Krishna	1,400 km	Mahabaleshwar (Maharashtra)	Tributaries: Koyna, Bhima, **Tungabhadra** (Tunga + Bhadra), Ghataprabha, Malaprabha, Musi (Hyderabad stands on the Musi); **Nagarjuna Sagar** and Srisailam dams, Almatti dam
Kaveri	800 km	Talakaveri, Brahmagiri hills (Karnataka)	Tributaries: Kabini, Hemavati, Bhavani, Amaravati; **Shivanasamudra falls** (site of one of Asia's earliest hydro-electric stations, 1902); Mettur dam (Stanley reservoir); delta in Tamil Nadu; Hogenakkal falls on its course
Pennar	597 km	Nandi hills (Karnataka)	Flows through AP to Nellore
Subarnarekha	395 km	Chotanagpur (Jharkhand)	Flows past Jamshedpur to the Bay of Bengal; **Hundru falls** on this river
Brahmani	800 km	Formed by Koel + Sankh rivers	Rourkela steel plant stands on its bank region (Odisha)
Vaigai	~258 km	Varusanadu hills (TN)	Madurai stands on its bank; ends in the Palk Bay
Tamraparni	~128 km	Agasthyamalai (TN)	Shortest major TN river; falls into the Gulf of Mannar

MEMORY HOOK: West-to-east/south order of the big deltas: Mahanadi (Odisha) → Godavari → Krishna (Andhra) → Kaveri (Tamil Nadu). Length order: **Godavari > Krishna > Narmada > Mahanadi > Kaveri**.

Peninsular rivers — west-flowing (Arabian Sea, estuary-forming)

West-flowing peninsular rivers are shorter, swift, and form **estuaries, not deltas**, because they flow through hard rock and steep rift valleys and the west coast is submerging.

River	Length	Key facts
Narmada	1,312 km	**Largest west-flowing peninsular river**; rises at **Amarkantak** (Maikal hills); flows west through a **rift valley** between the Vindhya (north) and Satpura (south); **Dhuandhar falls** at the marble rocks of Bhedaghat (Jabalpur); projects: **Sardar Sarovar** (Gujarat), Indira Sagar (MP); largest tributary: **Tawa**
Tapi (Tapti)	724 km	Rises at Multai (Betul, MP); flows west in a rift valley south of the Satpura (parallel to Narmada); Surat stands on it; Ukai dam; tributary Purna
Mahi	583 km	Rises in MP, flows through Rajasthan and Gujarat; **crosses the Tropic of Cancer twice**; Mahi Bajaj Sagar dam (Banswara)
Sabarmati	371 km	Rises in the Aravallis (Udaipur side); Ahmedabad and Gandhinagar stand on it; Dharoi dam
Luni	511 km	**Only major river of the Thar**; rises near Ajmer, becomes saline after Balotra, ends in the Rann of Kutch (inland drainage); Jawai is its main tributary
Sharavati	128 km	Karnataka river famous for the **Jog (Gersoppa) falls** — among India's highest waterfalls
Periyar	244 km	Kerala's longest river; Mullaperiyar dam (in Kerala, operated by Tamil Nadu — a standard inter-state fact)
Bharathapuzha	209 km	Kerala's second longest; drains the Pal Ghat gap region to the Arabian Sea
Mandovi and Zuari	Short	Goa's lifelines; Mandovi carries the **Dudhsagar falls**

EXAM TRAP: **Narmada and Tapi flow through rift valleys (graben)** — they did NOT carve their valleys by down-cutting like other rivers, which is why they flow west against the general eastward slope. The Narmada is longer than the Mahanadi and Kaveri; the Godavari still beats it overall.

PYQ PATTERN: "Which river crosses the Tropic of Cancer twice?" — **Mahi**. "Which is the largest river of the Indian desert?" — **Luni**. Statement sets: "Narmada flows through a rift valley" (true), "Tapi forms a large delta" (false — estuary).

Important waterfalls

Falls	River	State / note
Kunchikal	Varahi	Karnataka; **highest waterfall of India (about 455 m)**

Falls	River	State / note
Barehipani	Budhabalanga	Odisha (Similipal); among the tallest, two-tiered
Nohkalikai	Near Cherrapunji	Meghalaya; tallest plunge waterfall of India
Jog (Gersoppa/Mahatma Gandhi)	Sharavati	Karnataka; about 253 m, four cascades
Shivanasamudra	Kaveri	Karnataka
Hogenakkal	Kaveri	Tamil Nadu
Dhuandhar	Narmada	MP — smoke-like spray at Bhedaghat
Chitrakote	Indravati	Chhattisgarh; **widest** falls — "Niagara of India"
Hundru	Subarnarekha	Jharkhand (near Ranchi)
Dudhsagar	Mandovi	Goa

Lakes of India

Lakes are formed in many ways — tectonic movements (Wular), glacial action (most Himalayan lakes), river oxbow cut-offs (plains lakes), coastal lagoons (Chilika), volcanic craters (Lonar) and human dams (reservoirs). Exams test three things: the **state**, whether the water is **fresh or saline**, and one **special fact**. Learn them as a table.

Lake	State	Type	Special fact
Wular	Jammu & Kashmir	Fresh (tectonic)	**Largest freshwater lake of India**; the Jhelum flows through it
Dal	Jammu & Kashmir	Fresh	Srinagar's lake; floating gardens and houseboats
Manasbal	Jammu & Kashmir	Fresh	Among the deepest lakes of the Kashmir valley
Pangong Tso	Ladakh	Saline, endorheic	Very high-altitude lake; about one-third in India, rest in China; featured in films
Tso Moriri	Ladakh	Fresh-brackish	High-altitude Ramsar site in Changthang
Gurudongmar	Sikkim	Fresh (glacial)	One of the highest lakes of India (~5,183 m)
Cholamu (Tso Lhamo)	Sikkim	Fresh (glacial)	Often cited as India's highest lake (~5,330 m); source region of the Teesta
Tsomgo (Changu)	Sikkim	Fresh (glacial)	Near Nathu La route
Nainital, Bhimtal, Sattal	Uttarakhand	Fresh (glacial/tectonic)	Kumaun lake district
Roopkund	Uttarakhand	Fresh (glacial)	The "skeleton lake" of the high Himalaya

Lake	State	Type	Special fact
Loktak	Manipur	Fresh	**Largest freshwater lake in the North-East**; famous for floating **phumdis** and the Keibul Lamjao National Park — the world's only floating national park, home of the dancing deer (sangai)
Chilika	Odisha	Brackish lagoon	**Largest coastal lagoon / brackish water lake of India**; first Indian Ramsar site (1981); Nalabana bird sanctuary; Irrawaddy dolphins; separated from the sea by a sand bar
Pulicat	Andhra Pradesh–Tamil Nadu	Brackish lagoon	**Second largest lagoon of India**; Sriharikota island (Satish Dhawan Space Centre) sits in it; Nelapattu bird sanctuary
Kolleru	Andhra Pradesh	Fresh	Lies **between the Godavari and Krishna deltas**; famous pelican habitat (Atapaka)
Vembanad	Kerala	Brackish backwater	**Longest lake of India** (about 96 km); Nehru Trophy boat race (Punnamada, Alappuzha)
Ashtamudi	Kerala	Brackish backwater	Second largest wetland of Kerala; named for its eight channels
Sambhar	Rajasthan	Saline (inland)	**Largest inland salt lake of India**; major salt production; Ramsar site
Pushkar	Rajasthan	Fresh (natural sacred)	Holy lake at Ajmer side; Brahma temple town
Jaisamand (Dhebar)	Rajasthan	Fresh (artificial)	Among the largest artificial lakes, built by Maharana Jai Singh (17th century)
Pichola, Fateh Sagar, Nakki	Rajasthan	Fresh	Udaipur lake system; Nakki is at Mt. Abu
Lonar	Maharashtra	Saline-alkaline crater	Formed by a **meteor impact** in basalt rock; Ramsar site
Kanwar (Kabar)	Bihar	Fresh (oxbow)	**Largest oxbow lake of India** (Begusarai)
Deepor Beel	Assam	Fresh (oxbow/wetland)	Guwahati's Ramsar wetland
Nalsarovar	Gujarat	Fresh-shallow	Famous bird sanctuary near Ahmedabad
Hussain Sagar	Telangana	Fresh (artificial)	Links Hyderabad and Secunderabad

Lake	State	Type	Special fact
Bhojtal (Upper Lake, Bhopal)	Madhya Pradesh	Fresh (artificial)	11th-century lake attributed to Raja Bhoj; part of the Bhoj wetland
Renuka	Himachal Pradesh	Fresh	Largest natural lake of Himachal
Chandra Taal	Himachal Pradesh	Fresh (glacial)	Source region of the Chandra river (Spiti)

EXAM TRAP: India's largest **freshwater** lake is Wular; largest **saline/inland salt** lake is Sambhar; largest **lagoon/brackish** is Chilika; **longest** lake is Vembanad. Chilika is NOT a freshwater lake — a very common wrong statement. Keoladeo (Bharatpur) and Chilika were India's **first two Ramsar sites (1981)**; Keoladeo and Loktak are on the Montreux Record (wetlands under ecological stress).

MEMORY HOOK: Remember the four "largests" by one word each — **W**ular = fresh **W**ater, **S**ambhar = **S**alt, **Ch**ilika = **C**oastal, **V**embanad = longest (V for vast length).

PYQ PATTERN: Match-the-following frequently pairs Kanwar–oxbow, Lonar–crater, Chilika–lagoon, Wular–tectonic. Location traps: Pulicat straddles **AP and Tamil Nadu**; Pangong Tso is in **Ladakh**, not Himachal.

Irrigation in India

India has one of the largest irrigated areas in the world, yet only a little less than half of the net sown area is irrigated — the rest still depends on the monsoon. Irrigation matters because rainfall is seasonal, uneven and unreliable. There are three main sources, and their ranking is a fixed exam fact:

Wells and tube wells (over 55–60% of net irrigated area) > Canals (about 25–30%) > Tanks (about 3–5%), plus small "other" sources (ponds, check dams, streams).

Source	Where dominant	Why / notes
Wells and tube wells	UP, Punjab, Haryana, Rajasthan, Gujarat, Maharashtra, Tamil Nadu	Largest source; easy, private, on-demand; but causes groundwater over-use in Punjab–Haryana–Rajasthan; needs soft alluvium or weathered rock and power supply
Canals	Northern plains and coastal deltas (Punjab, Haryana, UP, AP, Tamil Nadu)	Need perennial rivers and gentle slopes; two types — inundation canals (no headworks, seasonal) and perennial canals (from dams/barrages)
Tanks	Peninsular India — Tamil Nadu, Andhra Pradesh, Karnataka, Telangana	Hard rocky surface will not absorb water, and the undulating land gives natural depressions to store rainwater; ancient tradition (eri system of Tamil Nadu)

- **Micro-irrigation** (drip and sprinkler) saves water and is promoted under **PMKSY — "Per Drop More Crop"**. Drip suits wide-spaced crops (fruits, cotton, sugarcane) and dry areas.

- Groundwater over-exploitation is worst in **Punjab, Haryana, Delhi and western UP/Rajasthan**, where the stage of groundwater development is above 100% in many blocks ("dark blocks").

Major irrigation and multipurpose projects

Project	River	States / note
Bhakra–Nangal	Satluj	Joint venture of Punjab, Haryana, Rajasthan; Bhakra dam (about 226 m) is among the world's highest **gravity dams**; its reservoir is **Gobind Sagar**; Jawaharlal Nehru described dams as the "temples of modern India" in the context of Bhakra
Hirakud	Mahanadi	Odisha; main dam 4.8 km with dykes totalling about 25.8 km — one of the **longest dams in the world**; among the first major projects after Independence
Damodar Valley (DVC)	Damodar	Jharkhand–West Bengal; 1948; first multipurpose project of independent India; dams: Tilaiya (first, on Barakar), Konar, Maithon, Panchet
Indira Gandhi Canal	From Harike barrage (Satluj–Beas confluence)	**Longest canal of India** (main canal about 649 km); earlier called the Rajasthan Canal; carries water to Ganganagar, Bikaner, Jaisalmer and Barmer districts of the Thar
Upper Ganga Canal	Ganga (Haridwar)	Opened 1854; historically among the earliest large modern canals
Nagarjuna Sagar	Krishna	Telangana–Andhra; one of the largest **masonry** dams; canals on both banks named after Jawaharlal Nehru and Lal Bahadur Shastri
Tehri	Bhagirathi	Uttarakhand; **highest dam of India** (about 260.5 m, rock-fill)
Sardar Sarovar	Narmada	Gujarat; terminal dam of the Narmada valley; benefits Gujarat, MP, Maharashtra, Rajasthan
Indira Sagar	Narmada	MP; among India's largest reservoirs by volume of storage
Chambal Project	Chambal	Joint MP–Rajasthan: **Gandhi Sagar → Rana Pratap Sagar → Jawahar Sagar → Kota Barrage** (in flow order)
Tungabhadra	Tungabhadra (Krishna system)	Shared by Karnataka and Andhra Pradesh
Kaleshwaram	Godavari (Telangana)	Described as the world's largest multi-stage **lift irrigation** project
Ken–Betwa Link	Ken → Betwa	**First river-interlinking project** taken up for implementation; benefits Bundelkhand (MP–UP)

EXAM TRAP: **Bhakra is a gravity dam**; **Tehri is the highest dam**; **Hirakud is among the longest dams**; the **Indira Gandhi Canal is the longest canal**. Nagarjuna Sagar's fame is "one of the world's largest masonry dams". Mixing highest/longest/largest across these four is the standard error the paper exploits.

PYQ PATTERN: Chambal project dam order (downstream) is a recurring ordering question: Gandhi Sagar (MP) → Rana Pratap Sagar (Rawatbhata) → Jawahar Sagar → Kota Barrage (all three later ones in Rajasthan).

Modes of irrigation — quick distinctions

- **Flow irrigation** — water flows by gravity from a higher source (canals, tanks in the plains).
- **Lift irrigation** — water is pumped up (wells, tube wells, river lifts like Kaleshwaram).
- **Inundation canals** — taken from rivers without a barrage; flow only when the river is in flood (old type, e.g., older canals of Punjab plains and Tamil Nadu deltas).
- **Warabandi** — the rotational system of distributing canal water by turns in north-western India (a favourite one-liner in agriculture questions).

Agriculture: Important Crops

Agriculture engages the largest share of India's workforce, and India has one of the world's largest cropped areas because of its varied climate — tropical in the south, temperate fruits in the Himalaya, desert farming in the west. About **45–46% of India's reported land area is under cultivation** (net sown area), one of the highest proportions in the world, against a world average near 11%. **Cropping intensity** is around 140% — the same field is cropped more than once because of irrigation and multiple cropping.

Cropping seasons

Season	Sowing	Harvest	Main crops
Kharif	June–July (with the SW monsoon)	September–October	Rice, jowar, bajra, maize, ragi, groundnut, cotton, jute, sugarcane, soybean, tur (arhar), moong, urad, sesame
Rabi	October–November (on stored moisture/irrigation)	April–May	Wheat, barley, gram, peas, mustard, linseed, potato
Zaid	March–June (short summer season)	Summer	Watermelon, muskmelon, cucumber, fodder crops, moong, vegetables

MEMORY HOOK: One-line memory — **Kharif comes with the rains, Rabi with the winter, Zaid fills the summer gap.** Rice is Kharif, wheat is Rabi — swapping them is the oldest trap in the book.

Types of farming (one-liners)

- **Primitive subsistence (shifting cultivation / slash and burn)** — called **Jhumming** in the North-East; other names: Bewar/Dahiya (MP), Podu/Penda (AP–Odisha), Kumari (Western Ghats, Kerala), Valre/Waltre (south-east Rajasthan), Khil (Himalayan belt), Kuruwa (Jharkhand), Pamlou (Manipur), Dipa (Bastar, Chhattisgarh).

- **Intensive subsistence farming** — small holdings, family labour, rice-dominant (densely populated monsoon Asia type).
- **Plantation agriculture** — single crop on large estates with capital and hired labour: tea, coffee, rubber, spices.
- **Commercial/dryland farming** — bajra, jowar, pulses and oilseeds in low-rainfall tracts; dryland farming is practised where annual rainfall is below about 75 cm.

The important crops — full table

Crop	Season	Climate/soil needs	Leading producer states	Key facts
Rice	Kharif	Hot and humid; 20–35°C; over 125 cm rain or assured irrigation; clayey alluvium that holds water	**West Bengal (largest)**, Punjab, UP, Tamil Nadu, Andhra Pradesh, Chhattisgarh	India is the **second largest producer** (after China) and a top exporter; three crops — **Aus, Aman, Boro** — in West Bengal/Assam/ Odisha; Chhattisgarh is called the **"Rice Bowl of India"**; Krishna–Godavari delta is a great rice tract
Wheat	Rabi	Cool, moist growing season (10–20°C) then warm dry ripening; well-drained loams	**Uttar Pradesh (largest)**, Madhya Pradesh, Punjab, Haryana	India is the 2nd largest producer (after China); the Green Revolution succeeded most in wheat (Punjab–Hary ana–western UP); HYV seeds came from **CIMMYT, Mexico** (Norman Borlaug; Indian architect: **M. S. Swaminathan**)
Jowar (sorghum)	Kharif (also rabi in Deccan)	Dry, warm; the great millet of the Deccan	**Maharashtra (largest)**, Karnataka, MP	Leading coarse grain by area among millets
Bajra (pearl millet)	Kharif	Hardy; thrives in hot, dry, sandy soils with 40–50 cm rain	**Rajasthan (largest)**, UP, Gujarat, Haryana	The classic crop of arid western India; survives where little else grows
Ragi (finger millet)	Kharif	Dry foot-hills and red soils	**Karnataka (largest)**, Tamil Nadu	Rich in calcium; Karnataka leads production
Maize	Kharif and rabi	Warm, moderate rain; grown in both seasons	Karnataka, Madhya Pradesh, Telangana, Bihar	Fastest-growing cereal; used for food, fodder and industry
Gram (chana)	Rabi	Cool, dry; light soils	**Madhya Pradesh (largest)**, Rajasthan, Maharashtra	**Most important pulse of India** by production
Tur (arhar/pigeon pea)	Kharif	Warm, moderate rain	Maharashtra, Karnataka, UP	Main kharif pulse

Crop	Season	Climate/soil needs	Leading producer states	Key facts
Pulses (overall)	Both	Mostly rain-fed dry tracts	Madhya Pradesh (largest overall)	India is the **largest producer, consumer and importer of pulses** in the world
Groundnut	Kharif	Light sandy loam; frost kills it	**Gujarat (largest)**, Rajasthan, Tamil Nadu	India is among the world's largest producers; Saurashtra is the core belt
Mustard/rapeseed	Rabi	Cool, dry; the oilseed of the north	**Rajasthan (largest)**, MP, Haryana, UP	Rajasthan leads mustard — a key Rajasthan fact
Soybean	Kharif	Warm, moderate rain; black soils	**Madhya Pradesh (~half of India's output — the "Soya State")**, Maharashtra	Introduced widely in the 1970s–80s; protein-rich oilseed
Sunflower	Both	Neutral; short duration	Karnataka (largest), Telangana/AP	Main rabi-kharif oilseed of the south
Cotton	Kharif	21–30°C; 50–100 cm rain; frost-free 210 days; **black (regur) soil** ideal	**Gujarat (largest)**, Maharashtra, Telangana	Long-staple (over 27 mm approx.) is premium; Bt cotton widely grown; India is among the world's top producers
Jute	Kharif	Hot, humid; 120–150 cm rain; new alluvium of flood plains	**West Bengal (dominant)**, Assam, Bihar	**"Golden fibre"**; Hooghly-side mills; India is the world's largest producer around with Bangladesh
Sugarcane	Annual (10–12 months)	Hot, humid tropic; 20–35°C; heavy rain/irrigation	**Uttar Pradesh (largest)**, Maharashtra, Karnataka	Sugar industry is India's **second largest agro-based industry** (after cotton textiles); peninsular cane has higher sucrose and a longer crushing season than northern cane, so mills are shifting south
Tea	Plantation	Warm, humid, frost-free; 150–300 cm rain well spread; sloping land with drainage; deep acidic soils	**Assam (largest — Brahmaputra valley)**, West Bengal (Darjeeling, Duars), Tamil Nadu (Nilgiris)	India is among the top producers and exporters; **Darjeeling tea was India's first GI-tagged product (2004)**; Tea Board HQ: Kolkata

Crop	Season	Climate/soil needs	Leading producer states	Key facts
Coffee	Plantation	Warm, humid hill slopes (600–1,600 m); shade trees	**Karnataka (about 70%)** — Kodagu, Chikkamagaluru, Hassan; Kerala, Tamil Nadu	Two species: **Arabica** (finer, higher altitude) and **Robusta**; legend of Baba Budan bringing beans from Arabia; Coffee Board HQ: Bengaluru
Natural rubber	Plantation	Hot, humid equatorial-type; over 200 cm rain; below 450 m in the classic belt	**Kerala (dominant)** — Kottayam belt; Tamil Nadu (Kanyakumari), Tripura	Rubber Board HQ: Kottayam
Spices	Various	Varied	Kerala — the **"Spice Garden of India"**	India is the world's largest producer, consumer and exporter of spices; black pepper and cardamom (Kerala–Karnataka), chillies (Andhra Pradesh — Guntur), turmeric (India is the largest producer; Telangana/Maharashtra/TN belts)

EXAM TRAP: Producer-state traps are endless — the safe anchors: **West Bengal = rice & jute; UP = wheat & sugarcane; Gujarat = groundnut & cotton; Rajasthan = bajra & mustard; MP = gram, soybean & pulses; Karnataka = coffee & ragi; Kerala = rubber & spices; Assam = tea; Maharashtra = jowar & sugar**.

Fibres, fruits and other headline facts

- India produces **all four natural silks — mulberry, tasar, eri and muga** — the only country to do so. **Muga** (golden silk) is unique to Assam (GI tagged). **Karnataka** is the largest silk producer (mulberry); the Central Silk Board is at Bengaluru.
- India is the **largest producer of milk** in the world (since 1998) — the result of **Operation Flood / the White Revolution** led by **Verghese Kurien** through NDDB, Anand (Gujarat).
- India is the **largest producer of mangoes and bananas** and the **second largest producer of fruits and vegetables overall** (after China). Mango: UP leads among states; grapes: Maharashtra (Nashik); apples: J&K and Himachal; litchi: Bihar (Muzaffarpur's Shahi litchi).
- **Rajasthan is the largest producer of wool** in India; Ludhiana (Punjab) is the hosiery hub.
- The **International Year of Millets was 2023**, declared by the UN on India's proposal; millets are promoted as **"Shri Anna"**.

Agricultural revolutions (match-the-following gold)

Revolution	Field	Associated name/fact
Green Revolution	Food grains (wheat, rice)	HYVP launched 1966–67; M. S. Swaminathan; Norman Borlaug

Revolution	Field	Associated name/fact
White Revolution	Milk	Operation Flood; Verghese Kurien; NDDB, Anand
Blue Revolution	Fish production	Inland and marine fisheries
Yellow Revolution	Oilseeds	Technology Mission on Oilseeds (1986)
Golden Revolution	Horticulture, honey	Fruits and vegetables boom period
Silver Revolution	Eggs/poultry	—
Red Revolution	Tomato/meat	—
Round Revolution	Potato	—
Grey Revolution	Fertilizers	—
Rainbow Revolution	All sectors together	—

PYQ PATTERN: Cropping-pattern statements to verify carefully — "Gram is a kharif pulse" is **wrong** (gram is rabi; tur is kharif). "Jute requires stagnant water/retting" — retting in water is the jute fibre process. "Sugarcane is a weight-losing raw material, so sugar mills locate near cane fields" — **correct** and explains mill geography.

Minerals: Metallic and Non-Metallic

India's mineral wealth sits mainly in the **old, hard rocks of the Peninsular plateau** — the shield areas of Chotanagpur, Odisha, Karnataka and Rajasthan. The northern alluvial plains are practically devoid of minerals, and petroleum is found in the sedimentary basins of Assam, Gujarat, Rajasthan and the offshore shelves. A useful map memory: minerals occur in **belts** — the North-Eastern Plateau belt (Chotanagpur–Odisha: coal, iron, mica, manganese, copper), the Central belt (Chhattisgarh–MP–AP), the Southern belt (Karnataka–Tamil Nadu), the South-Western belt (Karnataka–Goa coast: iron, garnet), the Western belt (Rajasthan–Gujarat: zinc, lead, marble, gypsum, copper), and the Himalayan belt (small deposits, plus petroleum in its foothill basins).

Metallic minerals

Iron ore

India has large, high-quality iron ore reserves. The two important ores are **hematite** (the main ore, "red ore", about 60–70% iron) and **magnetite** (the "black ore", up to about 70% iron, magnetic; concentrated in Karnataka). **Odisha is the largest producer of iron ore**, followed by Chhattisgarh, Karnataka and Jharkhand.

Belt / mine area	State	Fact
Keonjhar, Mayurbhanj, Sundargarh (Badampahar, Kiriburu, Meghahatuburu)	Odisha	Biggest hematite belt; feeds Rourkela and exports via Paradip
Bailadila (Dantewada)	Chhattisgarh	Among the world's best high-grade hematite; exported to Japan via Visakhapatnam port
Dalli-Rajhara	Chhattisgarh	Supplies ore to the **Bhilai steel plant**

Belt / mine area	State	Fact
Noamundi, Gua, Kiriburu (west Singhbhum)	Jharkhand	Oldest iron-ore mining belt; classic mine names
Ballari–Hospet, Sandur	Karnataka	Major producing belt
Kudremukh	Karnataka	Magnetite ore; name means "horse's face"
Goa mines	Goa	Lower-grade ore, almost entirely exported (via Mormugao)

Manganese

Essential for steel making (also batteries, chemicals). The classic belt is **Balaghat (Madhya Pradesh) – Nagpur/Bhandara (Maharashtra)**, with Odisha (Keonjhar–Sundargarh) holding the largest reserves; Karnataka (Shivamogga, Ballari) also produces. MOIL is the main public-sector producer. India was once the top exporter but now also imports high-grade ore — do not write "largest producer" for manganese.

Bauxite (aluminium ore)

- **Odisha is the largest producer (about half or more of India's output)** — the **Panchpatmali** deposit in Koraput district is among Asia's largest bauxite mines (NALCO).
- Other areas: Lohardaga (Jharkhand, called the "bauxite town"), Amarkantak–Maikal and Hasdeo (Chhattisgarh–MP), Kolhapur (Maharashtra), Kachchh/Jamnagar (Gujarat).
- Aluminium smelting needs cheap power, so plants sit near power sources, not just mines (Korba, Renukoot, Angul, Hirakud, Jharsuguda).

Copper

India is deficient in copper and imports concentrates. Three classic fields:

Field	State	Fact
Khetri–Singhana	Rajasthan (Jhunjhunu)	Most famous historic copper belt; Hindustan Copper
Singhbhum (Rakha, Mosabani, Surda)	Jharkhand	Old mining belt near Ghatsila
Malanjkhand	Madhya Pradesh (Balaghat)	**Largest copper mine of India** (open-cast)

Lead and zinc

Rajasthan dominates lead-zinc production through Hindustan Zinc: **Rampura-Agucha** (Bhilwara) — among the world's largest zinc-lead mines; also **Zawar** (Udaipur, one of the oldest zinc-mining areas), Rajpura-Dariba and Sindesar Khurd (Rajsamand). This is a high-value Rajasthan-specific fact for RAS.

Gold, silver and others

- **Kolar Gold Fields** (Karnataka) were among the deepest mines in the world; now closed. **Hutti** (Raichur, Karnataka) is the main working gold mine today; Ramagiri (Andhra Pradesh) is the other classic name.
- Silver is largely a by-product of lead-zinc (Rajasthan) and gold refining.
- Chromite: **Sukinda valley (Odisha)** holds the overwhelming bulk (about 95%+) of India's chromite.
- Nickel: Sukinda (Odisha) is also India's main nickel source.

- Tin: Chhattisgarh (Bastar) — India is poor in tin.

Fuel minerals — coal, petroleum, natural gas

Coal is India's most important fuel mineral (fuels the majority of electricity generation). About **98–99% of India's coal reserves are Gondwana coal** (bituminous, in the Damodar–Son–Mahanadi–Godavari valleys, about 200–300 million years old); the rest is **Tertiary coal** of the north-east.

Coal topic	Facts
Coal types (by carbon)	**Anthracite** (best, 80–95% carbon; small quantities in J&K) → **Bituminous** (60–80%; most Indian coal; Gondwana) → **Lignite** (brown coal, 40–55%; Neyveli in Tamil Nadu, also Rajasthan (Palana, Barmer) and Gujarat) → **Peat** (lowest grade)
Jharia	Jharkhand; **largest coalfield of India; the storehouse of prime coking coal**
Raniganj	West Bengal; **oldest coal mining area of India (mining began in 1774)**
Other Damodar valley fields	Bokaro, Giridih, Karanpura (Jharkhand–WB)
Other major fields	Talcher and Ib valley (Odisha — Talcher has among the largest reserves), Korba (Chhattisgarh), Singrauli (MP–UP), Singareni (Telangana, Godavari valley)
Tertiary coal	Makum (Assam), Namchik-Namphuk (Arunachal), Meghalaya, Nagaland fields
Position	India is the **second largest producer of coal in the world (after China)**; Coal India Ltd (HQ Kolkata) is the world's largest coal-producing company

- **Petroleum:** India's crude comes mainly from offshore and a few onland basins. **Mumbai High** (offshore, discovered 1974) is India's largest oilfield. **Digboi (Assam)** is the oldest oilfield, with India's first refinery (1901). Other fields: Naharkatiya, Moran (Assam); Ankleshwar, Kalol, and the Cambay basin (Gujarat); **Barmer basin (Rajasthan)** — the Mangala field is India's largest onland oil discovery of recent decades; Krishna–Godavari basin (gas-prone, offshore Andhra).
- **Jamnagar (Gujarat)** hosts the **world's largest oil refining complex** (Reliance). Other refining hubs: Digboi, Guwahati (Noonmati), Bongaigaon and Numaligarh (Assam); Koyali (Vadodara); Mathura; Panipat; Haldia; Mumbai; Chennai; Kochi; Visakhapatnam; Paradip; Mangaluru; Bathinda; Bina.
- India imports more than four-fifths of its crude oil requirement — a standard economy-geography link.
- **Natural gas:** major fields offshore Mumbai (Bassein), Hazira (Gujarat), KG basin; the **Hazira–Bijaipur–Jagdishpur (HBJ/HVJ)** pipeline carries gas to fertilizer and power plants of the interior.
- **Uranium and thorium:** Jaduguda (Jharkhand, Singhbhum) is India's pioneering uranium mine; large deposits also at Tummalapalle (Andhra Pradesh, Kadapa district). **Thorium** comes from the **monazite beach sands of Kerala** (and TN/Odisha coasts) — India has among the world's largest thorium reserves. Beach sands also yield ilmenite, rutile and zircon.

EXAM TRAP: Coal-type order is Anthracite > Bituminous > Lignite > Peat (by carbon and quality) — reverse order is the trap. "Largest coalfield" = Jharia; "largest coal reserves area" is often Talcher; "oldest" = Raniganj. Keep them separate.

Mica — the textbook special

India was historically the world's leading producer of sheet mica — the **Koderma–Giridih belt of Jharkhand** is called the **"mica capital of the world"**. Other areas: Nellore (Andhra Pradesh) and Bhilwara (Rajasthan). Mica's non-conducting property makes it vital for electrical insulation.

Non-metallic minerals

Mineral	Leading areas	Use / fact
Limestone	MP, Andhra Pradesh, Rajasthan, Chhattisgarh, Gujarat, Karnataka	Backbone of the **cement industry**; also steel flux and chemicals
Gypsum	**Rajasthan holds the bulk (about 80%+ of production)** — Bikaner, Nagaur, Jaisalmer, Barmer	Cement retarder, plaster of Paris, fertilizers
Rock salt	Mandi district (Drang and Guma mines), Himachal Pradesh; lake salt from Sambhar	India's rock-salt comes almost entirely from Mandi
Diamond	**Panna (Madhya Pradesh)** — India's **only** active diamond-mining area	Historical Golconda (Telangana–AP) gave the Koh-i-Noor and Hope diamonds' homeland regions
Kyanite	Jharkhand (Lapsa Buru, Singhbhum)	India has the **largest kyanite deposits in the world**
Barytes	**Mangampeta (Kadapa district, Andhra Pradesh)**	Among the world's largest single barytes deposits
Asbestos	Rajasthan (Udaipur), Andhra Pradesh	Fibrous insulation mineral
Feldspar, quartz, silica sand	Rajasthan (Ajmer area leads feldspar-quartz output) and others	Ceramics and glass
Steatite/talc (soapstone)	Rajasthan, Uttarakhand	Cosmetics, paints, paper filler
Marble	**Makrana (Rajasthan)** — the marble of the Taj Mahal; also Rajsamand, Udaipur	Dimension stones; Rajasthan leads in marble, granite (Jalore, "granite city")
Granite	Rajasthan, Tamil Nadu, Karnataka, AP	Polished stone exports
Phosphorite/rock phosphate	Rajasthan (Jhamarkotra, Udaipur), MP	Fertilizer input — a Rajasthan-linked fact
Fuller's earth (Multani mitti)	Rajasthan (Barmer, Bikaner, Jaisalmer)	Cosmetics and bleaching

MEMORY HOOK: Non-metallic Rajasthan cluster for RAS: **Gypsum–Rajasthan, Marble–Makrana, Rock phosphate–Jhamarkotra, Fuller's earth–Barmer, Lead-zinc–Rajasthan** — Rajasthan is India's mineral museum (it produces the widest variety of minerals among states, often summarised as the storehouse with about 80+ mineral types reported).

PYQ PATTERN: Classic match sets: Kudremukh–magnetite, Bailadila–hematite (iron); Khetri–copper; Koderma–mica; Panna–diamond; Sukinda–chromite; Neyveli–lignite; Jaduguda–uranium; Zawar–zinc. Statement trap: "India is the largest producer of coal" — **false**, China is first, India second.

Major Industrial Regions and Important Industries

Industries cluster where raw material, power, labour, transport and markets come together. India has **eight major industrial regions** (the standard textbook classification) plus several minor ones.

The eight major industrial regions

Region	Core centres	Main industries and why they are there
Mumbai–Pune	Mumbai, Thane, Pune	Cotton textiles (port for imports, humid climate, capital), petrochemicals, engineering, automobiles, films, IT; grew during the American Civil War cotton boom (1861–65)
Hugli (Kolkata–Hooghly)	Kolkata, Howrah, Hooghly belt	**Oldest industrial region**; jute (local raw jute initially, Hooghly water), engineering, paper, chemicals; riverine port + Damodar coal nearby + cheap labour
Bengaluru–Chennai	Bengaluru, Chennai, Salem link	Engineering, aircraft (HAL), electronics and IT in Bengaluru; automobiles, textiles, leather in Chennai belt; developed on local cotton and Pykara/Mettur hydel power
Gujarat	Ahmedabad–Vadodara–Surat	Cotton textiles (Ahmedabad — "Manchester of India"), petrochemicals (Jamnagar, Dahej, Hazira), chemicals, dairy (Amadavad–Anand, Amul), salt, diamond cutting (Surat)
Chotanagpur	Jamshedpur, Bokaro, Dhanbad, Ranchi	Heavy industries — iron and steel, heavy engineering, locomotives; sits directly on coal and iron ore; called India's heavy-industry heartland
Vishakhapatnam–Guntur	Visakhapatnam, Vijayawada, Guntur	Steel (Vizag plant), shipbuilding (Hindustan Shipyard), oil refining, petrochemicals, fertilizers, and agro-industries (tobacco, sugar) of the Krishna–Godavari delta
Gurugram–Delhi–Meerut	Delhi, Gurugram, Faridabad, Ghaziabad, Meerut	Light engineering, electronics, automobiles, software, garments, food processing — location driven by the capital's market, not raw material
Kollam–Thiruvananthapuram	Kollam, Kochi, Thiruvananthapuram, Alappuzha	Plantation processing (cashew — Kollam is the cashew capital), coir, fisheries, spices, plus Kochi's port, refinery and shipyard; powered by hydro-electricity

MEMORY HOOK: Region anchors — Mumbai = cotton+port, Hugli = jute+oldest, Chotanagpur = coal+steel, Gujarat = cotton+chemicals, Delhi belt = market, Kerala belt = plantation+hydel power.

Iron and steel industry

Steel is the backbone industry; its location is ruled by heavy, weight-losing raw materials (iron ore + coking coal), which is why most big plants hug the **Chotanagpur–Odisha** belt.

Plant	State	Fact
TISCO, Jamshedpur	Jharkhand	**First modern steel plant of India (1907, production from 1911–12)**; founded by Jamshedji Tata at Sakchi, on the Subarnarekha–Kharkai; private sector
IISCO, Burnpur	West Bengal	One of the pre-Independence trio (with VISL Bhadravati, 1923)
Bhilai	Chhattisgarh	**First public-sector plant of the Second Plan (1955), built with Soviet help**; Dalli-Rajhara ore, Korba power/coal region
Rourkela	Odisha	Built with **German** collaboration; first public-sector unit to start production (1959)
Durgapur	West Bengal	Built with **British** collaboration; on Damodar coal
Bokaro	Jharkhand	Built with **Soviet** help; among the largest plants
Salem	Tamil Nadu	Stainless/special steel
Visakhapatnam (RINL)	Andhra Pradesh	**Only shore-based (coastal) major steel plant of India**
Vijayanagara	Karnataka (Ballari)	Large private plant (JSW)

- **SAIL (Steel Authority of India Ltd, 1973)** runs the classic public-sector plants.
- India is the **world's second-largest producer of crude steel** (after China; it passed Japan in 2018) and the **largest producer of sponge iron (DRI)**.
- Collaboration trap: Bhilai & Bokaro = USSR; Rourkela = Germany (West Germany); Durgapur = UK (Britain).

EXAM TRAP: "First steel plant" questions — **Kulti (1874, Bengal Iron Works) made iron earlier**, but the first modern integrated steel plant in the PYQ sense is **TISCO Jamshedpur (1907)**. Visakhapatnam is shore-based; all other classic plants are inland, near raw materials.

Cotton textile industry

India's oldest and largest agro-based industry, giving the largest industrial employment after agriculture. The **first modern cotton mill was set up at Fort Gloster near Kolkata in 1818** (it failed); the first successful mill was the **Bombay Spinning and Weaving Mill (1854)** in Mumbai. Early growth favoured the west coast (raw cotton from the Deccan black-soil belt, port, humid climate that keeps threads from breaking, capital of traders).

- Leading states: Maharashtra, Gujarat, Tamil Nadu.
- Nicknames: **Ahmedabad — "Manchester of India"** (also called the Boston of the East); **Coimbatore — "Manchester of South India"**; **Kanpur — "Manchester of North India"**.
- Surat (synthetics), Mumbai, Indore, Nagpur, Solapur, Madurai, Salem, Ludhiana (hosiery) are other big centres.

Jute industry

- The **first jute mill was set up at Rishra (near Kolkata) in 1855**; the industry crowds both banks of the **Hooghly**.
- Partition (1947) split the industry: most mills stayed in India while most raw-jute land went to East Pakistan (now Bangladesh) — the classic textbook case of a political border breaking an industry.
- West Bengal dominates; National Jute Board HQ is Kolkata.

Sugar industry

- Sugar is a **weight-losing, seasonal** industry: cane loses sucrose quickly after cutting, so mills sit inside the cane belt and work only in the crushing season.
- **Uttar Pradesh has the largest number of sugar mills** and leads cane output; **Maharashtra** leads in cooperative sugar factories and sugar recovery.
- The industry has been **shifting towards the peninsula** (Maharashtra, Karnataka, TN) because tropical cane there has higher sucrose content, a longer crushing season and better-managed cooperatives.
- India is among the world's top two sugar producers (with Brazil).

Other important industries — one-line map

Industry	Headline facts
Cement	First plant attempted in 1904 (Madras/Chennai); first successful production 1914 (Porbandar); India is the **2nd largest producer** (after China); follows limestone belts — MP, AP/Telangana, Rajasthan, Chhattisgarh, TN; ACC, UltraTech are big private names
Paper and newsprint	First modern paper mill 1832 at Serampore (WB); first **newsprint mill at Nepanagar (MP)**; major centres in WB, AP (Rajahmundry), Karnataka (Bhadravati, Dandeli — bamboo), Odisha, Assam (bamboo-based), Gujarat
Fertilizers	First superphosphate factory 1906 at **Ranipet (TN)**; first big public-sector plant **Sindri (1951, Jharkhand, FCI)**; gas-based giants came up on the HBJ pipeline (Jagdishpur, Aonla, Phulpur); IFFCO plants at Kalol, Kandla, Phulpur, Aonla, Paradeep; India imports all its potash and much urea feedstock — "India is self-sufficient in potash" is FALSE
Aluminium	Smelters near cheap (hydro) power: Korba (BALCO), Renukoot (Hindalco), Hirakud, Jharsuguda, Angul (NALCO — its alumina refinery at Damanjodi and bauxite at Panchpatmali)
Copper smelting	Khetri complex (Rajasthan), Dahej (Gujarat), Ghatsila (Jharkhand's first copper smelter)
Automobiles	**Chennai — "Detroit of India"**; Pune and the Chakan belt; Gurugram–Manesar (Maruti Suzuki); Sanand (Gujarat); Pithampur (MP)
Information technology	**Bengaluru — "Silicon Valley of India"** and the IT capital; Hyderabad (Cyberabad), Pune, Chennai, Noida–Gurugram follow

Industry	Headline facts
Shipbuilding	Hindustan Shipyard, **Visakhapatnam (set up 1941)** — the first modern shipyard; Cochin Shipyard is the largest; Mazagon Dock (Mumbai), Garden Reach (Kolkata) build warships
Rail equipment	Locomotives: Chittaranjan (CLW) and Varanasi (DLW); coaches: ICF Perambur (Chennai), Rail Coach Factory Kapurthala; wheels: Yelahanka (Bengaluru)
Aircraft	Hindustan Aeronautics Ltd, **Bengaluru**(founded 1940)
Pharmaceuticals	India is called the **"pharmacy of the world"** — world's largest vaccine producer (Serum Institute, Pune); Hyderabad is a bulk-drug hub; Ahmedabad, Mumbai are other centres
Leather	Kanpur and Agra (footwear), Chennai–Vellore belt (Ambur, Ranipet), Kolkata
Silk	Karnataka leads (mulberry); Assam's muga, Jharkhand–Chhattisgarh tasar, eri in the North-East
Handloom/speciality towns	Firozabad — glass bangles; Moradabad — brassware; Aligarh — locks; Khurja — pottery; Panipat — handlooms ("city of weavers"); Sivakasi (TN) — matches and fireworks; Jalandhar and Meerut — sports goods; Surat — world's largest diamond cutting and polishing centre
Salt	Gujarat is the largest salt-producing state (about 70%+ of India's salt), then Tamil Nadu and Rajasthan (Sambhar)

PYQ PATTERN: Industry statement traps — "sugar mills locate near markets" is **false** (they locate near cane fields, weight-losing). "Steel plants are market-oriented" is generally **false** for India (raw-material oriented). "India is the largest producer of steel" is **false** (second). "Visakhapatnam steel plant is coal-based from local mines" — it is shore-based and imports much coking coal.

India Geography: Superlatives and One-Liners for Final Revision

Close the loop with the facts that appear as standalone one-mark questions. Read this list last, in the exam week.

- Largest state by area: **Rajasthan**; smallest: **Goa**; largest district: Kachchh.
- Longest river of India: **Ganga (2,525 km)**; longest peninsular river: **Godavari (1,465 km)**; largest west-flowing river: **Narmada (1,312 km)**.
- Longest tributary of the Ganga: **Yamuna**; largest by discharge: **Ghaghara**.
- River crossing the Tropic of Cancer twice: **Mahi**.
- Highest peak of India: **Kanchenjunga (8,598 m)**; highest entirely within India: **Nanda Devi (7,816 m)**; highest of South India: **Anamudi (2,695 m)**.
- Largest freshwater lake: **Wular**; largest salt lake (inland): **Sambhar**; largest lagoon: **Chilika**; longest lake: **Vembanad**; largest oxbow lake: **Kanwar**.
- Only active volcano of India: **Barren Island** (Andaman Sea).
- Channels: 8° (India–Maldives), 9° (Lakshadweep–Minicoy), 10° (Andaman–Nicobar).

- Wettest place on earth: **Mawsynram**; driest region: western Rajasthan (Thar) and the Ladakh cold desert.
- Monsoon: first onset on the Kerala coast, about **1 June**; about **75%** of annual rain falls June–September; average annual rainfall about **118 cm**.
- Largest delta in the world: **Ganga–Brahmaputra (Sundarbans)**; largest river island: **Majuli**.
- Highest waterfall: **Kunchikal (Karnataka)**; widest falls: **Chitrakote** on the Indravati.
- Largest canal: **Indira Gandhi Canal**; highest dam: **Tehri**; longest dam (with dykes): **Hirakud**; largest masonry dam among the classics: **Nagarjuna Sagar**; largest reservoir by storage among the big projects: Indira Sagar (Narmada).
- First multipurpose project: **Damodar Valley (1948)**; first river-linking project under implementation: **Ken–Betwa**.
- Biggest coalfield: **Jharia**; oldest coal mining: **Raniganj (1774)**; largest lignite field: **Neyveli**.
- Only diamond mines: **Panna (MP)**; largest copper mine: **Malanjkhand (MP)**; largest zinc-lead mine belt: **Rampura-Agucha (Rajasthan)**; mica capital: **Koderma (Jharkhand)**; bauxite giant: **Panchpatmali (Odisha)**; chromite: **Sukinda (Odisha)**; uranium pioneer: **Jaduguda (Jharkhand)**; thorium: **Kerala monazite sands**.
- Only country producing all four natural silks: **India**; largest silk state: **Karnataka**; largest wool state: **Rajasthan**.
- Manchester of India: **Ahmedabad**; of South India: **Coimbatore**; Detroit of India: **Chennai**; Silicon Valley of India: **Bengaluru**.
- First GI tag of India: **Darjeeling tea (2004)**.
- Largest milk, mango and banana producer country in the world: **India**; largest producer of pulses and spices in the world: **India**; largest producer-consumer-importer of pulses: **India**.
- Longest coastline state: **Gujarat**; longest border: with **Bangladesh**; standard meridian: **82°30′E (Mirzapur)**.

MEMORY HOOK: Final check trick — for every superlative ask "largest by WHAT?" (area/length/production/discharge). RAS traps live inside that single question: Yamuna is longest, Ghaghara is largest; Wular is freshest-biggest, Chilika is biggest but brackish; Jharia is biggest field, Talcher holds the biggest reserves.